

Chapter 9: Determinants versus Row Operations Worksheet

1. Are each of the following statements true or false?

- (a) If A is a 3×3 matrix with $\det(A) = 3$, then $\det(2A) = 6$.
- (b) If A is a 4×4 matrix with row 2 = $[3 \quad 5 \quad -1 \quad 0]$ and row 4 = $[-9 \quad -15 \quad 3 \quad 0]$, then $\det(A) = 0$.
- (c) For any matrix A , A and $-A$ will have the same determinant.

2. Suppose A, B are 4×4 matrices, and B is obtained from A by the following row operations:

Swap r_2 and r_3

Multiply r_2 by $\frac{1}{5}$

Add $-3r_1$ to r_2 (and replace r_2)

Add $8r_3$ to r_4 (and replace r_4)

Multiply r_3 by 6

If $\det(B) = -10$, find $\det(A)$.

3. Matrix C was reduced to $\begin{bmatrix} 4 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 \\ 4 & 2 & -10 & 0 \\ 0 & 3 & 5 & 5 \end{bmatrix}$ via 43 row replacements (i.e. add cr_i to r_j with $i \neq j$), a row scale by 20, and 27 row swaps (i.e. swap r_i with r_j , $i \neq j$). Find $\det(C)$.

4. Compute the following determinants using elementary row operations to reduce the matrix to an easily computable determinant.

(a) $\begin{vmatrix} 4 & -3 & 5 \\ 5 & 2 & 0 \\ 2 & 0 & 4 \end{vmatrix}.$

(b) $\begin{vmatrix} 4 & 1 & 3 \\ 2 & 3 & 0 \\ 1 & 3 & 2 \end{vmatrix}.$

(c) $\begin{vmatrix} 2 & 3 & 1 \\ 3 & 5 & 2 \\ 1 & 2 & 1 \end{vmatrix}.$